Parallel Computing Benchmark Document — Page 1 of 5

Amdahl's Law and Theoretical Speedup Bounds (Section 1)

Amdahl's law is a formula which gives the theoretical speedup in latency of the execution of a task at fixed workload that can be expected of a system whose resources are improved. Named after computer scientist Gene Amdahl, it states that if a program has a serial portion s, the maximum achievable speedup using p processors is bounded by 1 / (s + (1 - s) / p). Even with an infinite number of processors, the speedup can never exceed 1 / s.

Process Scheduling and Concurrency in Operating Systems (Section 2)

Concurrency is the execution of multiple instruction sequences at the same time. In modern operating systems, multiprocessing allows true parallel execution on multiple CPU cores, avoiding the limitations of the Global Interpreter Lock (GIL) found in CPython. When designing parallel algorithms, worker allocation and process creation cost must be carefully evaluated to avoid performance degradation on small computational jobs.

Data Partitioning and Load Balancing (Section 3)

In distributed and shared-memory parallel computing, load balancing refers to the practice of distributing work evenly across available computing resources. A parallel efficiency metric measures the fraction of time for which processors are being used constructively. Imbalanced partitions cause worker starvation where fast workers idle while waiting for the slowest.

Inter-Process Communication and Overhead (Section 4)

Inter-process communication (IPC) mechanisms allow processes to manage shared data. However, excessive serialization (pickling) and IPC bandwidth can become a significant bottleneck. In our PDF extractor design, each worker independently reads its assigned page range from disk, eliminating redundant memory transfers across process boundaries and boosting throughput.

Parallel Computing Benchmark Document — Page 2 of 5

Process Scheduling and Concurrency in Operating Systems (Section 1)

Concurrency is the execution of multiple instruction sequences at the same time. In modern operating systems, multiprocessing allows true parallel execution on multiple CPU cores, avoiding the limitations of the Global Interpreter Lock (GIL) found in CPython. When designing parallel algorithms, worker allocation and process creation cost must be carefully evaluated to avoid performance degradation on small computational jobs. Notice: The keyword 'Amdahl' is discussed in detail within Section 1 regarding system scaling and parallel efficiency benchmarks.

Data Partitioning and Load Balancing (Section 2)

In distributed and shared-memory parallel computing, load balancing refers to the practice of distributing work evenly across available computing resources. A parallel efficiency metric measures the fraction of time for which processors are being used constructively. Imbalanced partitions cause worker starvation where fast workers idle while waiting for the slowest. Notice: The keyword 'Amdahl' is discussed in detail within Section 2 regarding system scaling and parallel efficiency benchmarks.

Inter-Process Communication and Overhead (Section 3)

Inter-process communication (IPC) mechanisms allow processes to manage shared data. However, excessive serialization (pickling) and IPC bandwidth can become a significant bottleneck. In our PDF extractor design, each worker independently reads its assigned page range from disk, eliminating redundant memory transfers across process boundaries and boosting throughput. Notice: The keyword 'Amdahl' is discussed in detail within Section 3 regarding system scaling and parallel efficiency benchmarks.

Parallel Computing and High Performance Systems (Section 4)

Parallel computing is a type of computation in which many calculations or processes are carried out simultaneously. Large problems can often be divided into smaller ones, which can then be solved at the same time. There are several different forms of parallel computing: bit-level, instruction-level, data, and task parallelism. In modern multi-core architectures, achieving maximum speedup requires minimizing synchronization overhead and balanced workload distribution among active worker processes. Notice: The keyword 'Amdahl' is discussed in detail within Section 4 regarding system scaling and parallel efficiency benchmarks.

Parallel Computing Benchmark Document — Page 3 of 5

Data Partitioning and Load Balancing (Section 1)

In distributed and shared-memory parallel computing, load balancing refers to the practice of distributing work evenly across available computing resources. A parallel efficiency metric measures the fraction of time for which processors are being used constructively. Imbalanced partitions cause worker starvation where fast workers idle while waiting for the slowest. Notice: The keyword 'Amdahl' is discussed in detail within Section 1 regarding system scaling and parallel efficiency benchmarks.

Inter-Process Communication and Overhead (Section 2)

Inter-process communication (IPC) mechanisms allow processes to manage shared data. However, excessive serialization (pickling) and IPC bandwidth can become a significant bottleneck. In our PDF extractor design, each worker independently reads its assigned page range from disk, eliminating redundant memory transfers across process boundaries and boosting throughput. Notice: The keyword 'Amdahl' is discussed in detail within Section 2 regarding system scaling and parallel efficiency benchmarks.

Parallel Computing and High Performance Systems (Section 3)

Parallel computing is a type of computation in which many calculations or processes are carried out simultaneously. Large problems can often be divided into smaller ones, which can then be solved at the same time. There are several different forms of parallel computing: bit-level, instruction-level, data, and task parallelism. In modern multi-core architectures, achieving maximum speedup requires minimizing synchronization overhead and balanced workload distribution among active worker processes. Notice: The keyword 'Amdahl' is discussed in detail within Section 3 regarding system scaling and parallel efficiency benchmarks.

Amdahl's Law and Theoretical Speedup Bounds (Section 4)

Amdahl's law is a formula which gives the theoretical speedup in latency of the execution of a task at fixed workload that can be expected of a system whose resources are improved. Named after computer scientist Gene Amdahl, it states that if a program has a serial portion s, the maximum achievable speedup using p processors is bounded by 1 / (s + (1 - s) / p). Even with an infinite number of processors, the speedup can never exceed 1 / s. Notice: The keyword 'Amdahl' is discussed in detail within Section 4 regarding system scaling and parallel efficiency benchmarks.

Parallel Computing Benchmark Document — Page 4 of 5

Inter-Process Communication and Overhead (Section 1)

Inter-process communication (IPC) mechanisms allow processes to manage shared data. However, excessive serialization (pickling) and IPC bandwidth can become a significant bottleneck. In our PDF extractor design, each worker independently reads its assigned page range from disk, eliminating redundant memory transfers across process boundaries and boosting throughput. Notice: The keyword 'Amdahl' is discussed in detail within Section 1 regarding system scaling and parallel efficiency benchmarks.

Parallel Computing and High Performance Systems (Section 2)

Parallel computing is a type of computation in which many calculations or processes are carried out simultaneously. Large problems can often be divided into smaller ones, which can then be solved at the same time. There are several different forms of parallel computing: bit-level, instruction-level, data, and task parallelism. In modern multi-core architectures, achieving maximum speedup requires minimizing synchronization overhead and balanced workload distribution among active worker processes. Notice: The keyword 'Amdahl' is discussed in detail within Section 2 regarding system scaling and parallel efficiency benchmarks.

Amdahl's Law and Theoretical Speedup Bounds (Section 3)

Amdahl's law is a formula which gives the theoretical speedup in latency of the execution of a task at fixed workload that can be expected of a system whose resources are improved. Named after computer scientist Gene Amdahl, it states that if a program has a serial portion s, the maximum achievable speedup using p processors is bounded by 1 / (s + (1 - s) / p). Even with an infinite number of processors, the speedup can never exceed 1 / s. Notice: The keyword 'Amdahl' is discussed in detail within Section 3 regarding system scaling and parallel efficiency benchmarks.

Process Scheduling and Concurrency in Operating Systems (Section 4)

Concurrency is the execution of multiple instruction sequences at the same time. In modern operating systems, multiprocessing allows true parallel execution on multiple CPU cores, avoiding the limitations of the Global Interpreter Lock (GIL) found in CPython. When designing parallel algorithms, worker allocation and process creation cost must be carefully evaluated to avoid performance degradation on small computational jobs. Notice: The keyword 'Amdahl' is discussed in detail within Section 4 regarding system scaling and parallel efficiency benchmarks.

Parallel Computing Benchmark Document — Page 5 of 5

Parallel Computing and High Performance Systems (Section 1)

Parallel computing is a type of computation in which many calculations or processes are carried out simultaneously. Large problems can often be divided into smaller ones, which can then be solved at the same time. There are several different forms of parallel computing: bit-level, instruction-level, data, and task parallelism. In modern multi-core architectures, achieving maximum speedup requires minimizing synchronization overhead and balanced workload distribution among active worker processes.

Amdahl's Law and Theoretical Speedup Bounds (Section 2)

Amdahl's law is a formula which gives the theoretical speedup in latency of the execution of a task at fixed workload that can be expected of a system whose resources are improved. Named after computer scientist Gene Amdahl, it states that if a program has a serial portion s, the maximum achievable speedup using p processors is bounded by 1 / (s + (1 - s) / p). Even with an infinite number of processors, the speedup can never exceed 1 / s.

Process Scheduling and Concurrency in Operating Systems (Section 3)

Concurrency is the execution of multiple instruction sequences at the same time. In modern operating systems, multiprocessing allows true parallel execution on multiple CPU cores, avoiding the limitations of the Global Interpreter Lock (GIL) found in CPython. When designing parallel algorithms, worker allocation and process creation cost must be carefully evaluated to avoid performance degradation on small computational jobs.

Data Partitioning and Load Balancing (Section 4)

In distributed and shared-memory parallel computing, load balancing refers to the practice of distributing work evenly across available computing resources. A parallel efficiency metric measures the fraction of time for which processors are being used constructively. Imbalanced partitions cause worker starvation where fast workers idle while waiting for the slowest.